INTELLIGENT BRAKE MONITOR

TOPIC: A small electronic monitor that watches a brake's temperature and how hard it's being pressed, and tells you whether the brake is behaving normally or showing early signs of a problem. When brakes get too hot from repeated hard use, they can lose stopping power also called brake fade. Right now, most brakes don't warn us as to why this is happening; you just notice the pedal feels different. Your project is a tool that would tell a driver *before* the problem becomes dangerous.

REQUIRED COMPONENTS

- A temperature sensor- checks how hot the brake is getting
- A force sensor - checks how hard the brake is being pressed
- OLED display - shows the driver a simple status: is everything OK, or is something concerning happening

Intelligent Brake Health Monitor — Project Summary

What this project is

A small electronic device that watches a brake's **temperature**, **applied force**, and **vibration**, and tells you in real time whether things look Normal, Elevated, at risk of Fade, or Critical — before a real fade problem becomes dangerous.

Why temperature + force + vibration together

- **Temperature alone isn't enough** — brakes are supposed to get warm. What matters is whether it's heating up *too fast*, or getting *too hot*.
- **Force tells you if the heat matches the effort** — if temperature is climbing fast while force stays normal, that's the actual signature of brake fade (the pad losing grip and needing more heat/effort to do the same job).
- **Vibration catches a different problem** — a warped or unbalanced rotor, separate from heat issues.

Components used

Component	What it senses	How it connects
LM35	Temperature	Analog (A0)
Interlink FSR 402	Force	Analog (A1), with a 10kΩ resistor forming a voltage divider
MPU6050	Vibration (via acceleration)	I2C (SDA/SCL)
Arduino Uno	Runs all the logic	—

How the values are calculated

- **Temperature**: LM35 outputs 10mV per °C, so `tempC = voltage * 100`
- **Temperature rate of change**: compares the current reading to the reading from the last loop, divided by time elapsed — tells us how many °C per second it's climbing
- **Force**: raw FSR reading converted to Newtons using a resistance/conductance formula, then corrected with a calibration factor based on a real 10N test (the generic formula alone was inaccurate for this specific sensor)
- **Vibration**: MPU6050 gives acceleration on X/Y/Z axes; combined into one magnitude number, then gravity's constant ~9.8 m/s² is subtracted so only "extra" shaking remains

Status logic (what decides Normal/Elevated/Fade/Critical)

1. **CRITICAL** — temperature ≥ 300°C
2. **HIGH VIBRATION** — vibration above threshold (currently being calibrated against real resting baseline)
3. **FADE RATE RISK** — temperature rising fast AND force is still adequate (the actual fade signature)
4. **HIGH RISK TEMPERATURE** — temperature between 250-300°C
5. **HIGH TEMPERATURE** — temperature between 150-250°C
6. **NORMAL** — everything else

Testing approach (since no full brake rig was built)

- **Temperature**: tested using a heat source near the LM35, not a real rotor
- **Force**: tested by pressing the FSR with known weights to calibrate it
- **Vibration**: tested by shaking/moving the MPU6050 directly

This means the system proves the *sensing and decision logic* works correctly — it's not yet validated against a real spinning brake rotor and caliper.

Current status

- All three sensors wired and confirmed working individually and together
- Full decision logic implemented and printing to Serial Monitor
- Force calibrated to real Newtons using a known-weight test
- Vibration threshold still being tuned against a true "resting" baseline
- OLED display and data logging (SD card) are planned next steps, not yet added

What's next

1. Finish calibrating the vibration threshold (test resting baseline vs. real vibration)
2. Add the OLED display to show live values + status
3. Mount everything (CAD enclosures, sensor brackets)
4. Add SD card data logging (later addition)

Developing an embedded brake health monitoring system that uses temperature, applied force, and vibration measurements to evaluate brake performance and identify potential issues. The system collects sensor data during braking and compares it with baseline conditions and defined thresholds to detect changes in brake behavior. Based on the sensor measurements, the system classifies brake conditions as Normal, High Temperature, High-Risk Temperature, Fade Risk, High Vibration, or Critical. The project combines PCB development, sensor integration, embedded programming, signal processing, and mechanical design to develop an early-warning system for potential brake fade and mechanical degradation.